

When Does Filtering Help You See a Break? Latent-State Diagnostics for Structural Change at Calibrated False-Alarm Rates

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Abstract

A two-layer framework for detecting hidden structural change: a state-estimation layer (Kalman filtering of a latent state from noisy observations) and a diagnostics layer that converts features of the strictly-causal filtered path into alarms. All detectors — ours and benchmarks — are calibrated on matched null data-generating processes to the same false-alarm rate (5% per 500 observations), making detection rates and delays directly comparable. The main finding is a division of labor, not a victory: for level shifts, a CUSUM on the raw data dominates detection rates at every signal-to-noise ratio, while the latent-innovation CUSUM is “fast or never” — a phenomenon we formalize (the post-break innovation mean decays geometrically to $\delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F})$; when this is below the CUSUM allowance, post-transient detection has an exponentially small bound) and verify numerically — sweeping the state persistence φ , we find the asymptotic drift μ_∞ sorts detection (Spearman 0.94) and the innovation CUSUM *escapes* the fast-or-never regime at low φ , so the negative result is a boundary condition of persistent states (the empirically relevant case), not a universal law. For second-moment changes the story is a two-channel one. On observation-noise breaks a whitening ladder shows the edge over raw data is *prewhitening under autocorrelation*, not the latent state estimate: the ARIMA-residual and Kalman-innovation rungs are provably the *same filter* (the observable is exactly ARMA(1,1), so the two whitened rungs coincide up to estimation error, $\bar{\rho} = 0.99$), and both detect the subtle ($\times 1.5$) break at every SNR (ARIMA 0.90/0.94/0.87) while a raw-data variance CUSUM detects it only when observation noise dominates (0.996 at SNR 0.1) and falls to chance as the latent signal grows (0.56, 0.10 at SNR 0.5, 2.0). But this advantage is specific to the observation-noise channel: on state-innovation (shock-variance) breaks — the channel that describes the Great Moderation and crisis-volatility events the application targets — the ordering *inverts*, and a raw variance CUSUM matches or beats the whitened rungs at every SNR ($\times 1.5$: raw 0.09/0.21/0.23 vs ARIMA 0.03/0.10/0.16). Prewhitening reveals a break in the white component and *removes* the signal of a break in the state, which is why on real crises a raw variance CUSUM’s timing is indistinguishable from the state-aware composite’s. The diagnostics further survive heavy tails via an exceedance-indicator variant and, via a shortfall CUSUM, detect variance *quieting* that level-oriented methods miss. A real-data application (industrial production, GDP, Treasury yields) reproduces the profile: every alarm attributes to a second-moment feature, association with NBER reference dates is significant (permutation $p = 0.007$), and real-time (ALFRED vintage) analysis confirms the COVID timing while honestly downgrading the 2008 timing claim.

JEL classification: C12, C22, C52. Keywords: structural change; sequential change detection; CUSUM; state-space models; Kalman filter; prewhitening; false-alarm rate; latent-state diagnostics; volatility

regimes.

1. Introduction

Structural change is usually sought in observables. But many economic quantities of interest — trend output, a natural rate, an underlying volatility regime — are latent states observed only with noise, and the natural first step is to filter: estimate the state, then watch the estimate for breaks. Intuition says the filtering should help, by stripping observation noise before the change is sought. This paper asks when that intuition holds, under a protocol that makes the question answerable. Our contributions are as follows.

- Contribution 1 (protocol): a calibrated-FAR parity harness. Every method's threshold comes from the same routine on the same matched-null draws; empirical FAR is re-verified on fresh nulls; causality is enforced structurally (parameters fit on a training prefix only; forward-only filtering; alarm scores are NaN on training data) and tested bit-identically (perturbing future observations leaves all scores at earlier times unchanged, exactly).
- Contribution 2 (negative + positive results): the intuition is wrong for first moments at high persistence — and we can prove why (fast-or-never theorem), then bound the claim by sweeping φ ($\mu\infty$ sorts detection, and the innovation CUSUM escapes the regime at low φ). For second moments the answer is a two-rung, two-channel decomposition. The ladder collapses to *two* rungs because the ARIMA and Kalman rungs are the same filter — the observable is exactly ARMA(1,1), an equivalence we state as theory and confirm to machine precision — so "prewhitening" and "state estimation" are not separable here; the state layer adds nothing the ARIMA residuals do not. And the prewhitening advantage is *channel-specific*: it holds for observation-noise breaks (decisive at high SNR, where the latent signal masks the noise change and raw sits at chance) but *inverts* for state-innovation breaks, where the raw variance CUSUM matches or beats the whitened rungs — the channel that matters for the volatility-regime events of the application.
- Contribution 3 (method): a tail-robust exceedance-indicator CUSUM that preserves the second-moment advantage under heavy-tailed noise, reached by rejecting two documented alternatives — a falsified clipped variant and an in-composite variant diluted below its standalone form (§8.3).
- Contribution 4 (application discipline): attribution, permutation tests, sensitivity, pinned data snapshots, and real-time vintages for the real-data claims — including a self-correction on the headline GFC timing.
- Honest-outcome framing throughout: three pre-registered hypotheses were falsified; every post-hoc change is logged (CHANGELOG) and the failures are reported as findings.

Related work. The paper sits at the intersection of several literatures. Sequential change detection descends from Page's (1954) CUSUM and the quickest-detection tradition (Lorden 1971; Moustakides 1986); we use the CUSUM as the common statistic across information sets rather than proposing a new stopping rule. The idea of monitoring a *state-space* model through its innovations is not ours: it is the core of the innovation-based change-detection literature — the generalized likelihood ratio test on Kalman innovations (Willsky & Jones 1976) and the systematic treatment of CUSUM/GLR schemes on innovation sequences in Basseville & Nikiforov (1993). That literature also supplies the evaluation currency we adopt: the average run length (ARL), with methods compared at a matched in-control ARL_0 — the statistical-process-control convention (Page 1954; Montgomery). Our contribution on this axis is *not* the innovation CUSUM or ARL-matching per se, both of which are standard in SPC and quickest detection; it is that this matched-error protocol is essentially *absent* from the applied latent-state econo-

metrics literature, and that transplanting it there — calibrating every information set to a common false-alarm rate — is precisely what produces the negative results (raw data wins on levels; the state estimate adds nothing over ARIMA whitening on variances). In econometrics, sequential monitoring of structural change is the CUSUM-of-recursive-residuals line and its modern monitoring form (Brown, Durbin & Evans 1975; Chu, Stinchcombe & White 1996), which watch *observable* regression residuals; our object is a *latent* state, and the residuals we monitor are a filter's innovations. Regime-switching models (Hamilton 1989; Kim–Nelson 1999) offer a state-aware alternative whose regime probabilities we include as a benchmark and find saturate under calibration on nonstationary data. The empirical target of the second-moment results is the Great Moderation volatility decline (McConnell–Perez-Quiros 2000; Stock–Watson 2002); as we show, that and the crisis-volatility events are *state-innovation* (shock-variance) breaks, the channel on which raw and prewhitened detectors are interchangeable. Finally, offline changepoint methods (PELT, Killick et al. 2012) solve a retrospective segmentation problem; our monitoring is strictly causal and calibrated to a false-alarm rate. What is new here is not any single detector but the calibrated-parity harness that makes latent-state and raw-data detectors directly comparable, and a reduced-form result (the exact ARMA(1,1) equivalence of the ARIMA and Kalman rungs) that dissolves the “does filtering help?” question for second moments into a prewhitening question with a known answer.

2. Framework and evaluation protocol

Model layer. $S_t = \phi S_{t-1} + w_t$ (var q), $Y_t = S_t + v_t$ (var r); the estimator sees only Y and fits (ϕ, q, r) by maximum likelihood on a training prefix (25% of the sample), then runs a forward-only filter with frozen parameters. Standardized one-step innovations e_t and the filtered state are the raw material for diagnostics.

Diagnostics layer. Eleven features of the filtered path — level change, slope, acceleration, instability, rolling persistence, a Page CUSUM of innovations (allowance $k = 0.5$), variance CUSUMs of $e^2 - 1$ ($k = 0.25$ and 0.05), a quietness CUSUM of $1 - e^2$ ($k = 0.05$), rolling innovation autocorrelation, and a CUSUM of the filtered state against its training baseline. Each feature is standardized at every time point by the median and IQR of its null distribution at that same time point (pooled-over-time standardization is a design flaw that blunted the composite; §8.4). The composite score is the max of standardized features; the composite is itself calibrated on nulls, which prices in the multiple-feature testing automatically.

Calibration parity. Threshold = $(1 - \text{FAR})$ quantile of the per-replication maximum score over matched-null draws; identical routine, budgets, and seed layout for LSC detectors and benchmarks (raw- Y CUSUM, ARIMA+CUSUM, plain-HMM regime flips). Calibration, evaluation, FAR-check, and feature-scale seeds are disjoint by construction.

Metrics. Detection rate, pre-break false alarms, censored delay; for multi-break: event-level precision/recall/F1 with one-to-one greedy matching in a 100-observation window. To connect the protocol to the statistical-process-control and quickest-detection literatures we also report the two average-run-length quantities those fields are built around (`paper_assets/ar1_table.csv`). The in-control ARL_0 is the mean observations between false alarms implied by the calibrated per-observation false-alarm hazard: with monitored window length $L = 375$ ($T = 500$, 25% training) and empirical window-FAR α , the hazard is $p = 1 - (1 - \alpha)^{1/L}$ and $\text{ARL}_0 = 1/p \approx L/\alpha$; our 5% window target corresponds to $\text{ARL}_0 \approx 7300$ observations, and the empirical detectors sit at 5900–7600 (raw CUSUM calibrates slightly hot). The out-of-control ARL_1 is the post-break detection delay (mean delay conditional on detection, reported beside the detection rate since misses are censored): e.g. at 3σ level, SNR 0.5, $\text{ARL}_1 \approx 77\text{--}86$

observations. Calibrating a common ARL_0 is exactly the ARL-matching convention of SPC (Basseville & Nikiforov 1993); we express it as a window-FAR because our horizon is finite.

3. Simulation design

Arenas: AR(1) latent state ($\varphi = 0.95$) at spec-SNR (stationary state variance / observation variance) $\in \{0.1, 0.5, 2.0\}$, with a persistence sweep $\varphi \in \{0.5, 0.8, 0.95, 0.99\}$ (SNR held fixed by $q = \text{SNR} \cdot (1 - \varphi^2)$) used to map the fast-or-never boundary (§4); and a local-level (random-walk state) arena, analyzed in §8.4, where — as we now demonstrate rather than assert — level detection is degenerate for every method and variance detection becomes possible only after whitening. Breaks at mid-sample come in two variance channels whose distinction is central to §5. An *observation-noise* (r) break scales the white-noise component's standard deviation ($\times 1.5, \times 3, \times 2/3$ quieting); a *state-innovation* (q) break scales the state shock's standard deviation by the same factors — the identical SD convention, so " $\times 1.5$ " means $SD \times 1.5$ in both channels. The two are structurally different: an r-break changes only the marginal variance of Y, whereas a q-break changes both the marginal variance and the autocorrelation structure (it shifts the reduced-form ARMA(1,1) MA parameter). Also: level shifts ($0.5, 1, 3 \sigma_{\text{ref}}$), logistic ramps, and pure persistence changes ($\varphi \rightarrow 0.995$ or 0.80 with the stationary variance held fixed). $T = 500$ baseline with a $T \in \{200, 2000\}$ sweep; misspecification arenas with t_5 observation noise and a nonlinear tanh state drift.

4. First moments: raw data wins, and we can say exactly why

Empirics. At matched FAR, raw-Y CUSUM has the best level-shift detection rate at every SNR ($0.97 - 0.99$ at 3σ) and every T; the latent state CUSUM approaches but never overtakes it ($0.19 \rightarrow 0.94$ as SNR rises). Two pre-registered "latent advantage" hypotheses were falsified. The latent-innovation CUSUM is the speed champion conditional on firing: median delay 24–53 observations at 3σ versus raw's 58–91, at the cost of detection rate ($0.55 - 0.67$) — "fast or never."

Theory. The empirics above are not an artifact of tuning; they follow from two facts about the detectors' post-break drift. Derivations are in Appendix B (`experiments/THEORY.md`).

Proposition 1 (fast-or-never). After a state level shift δ , the standardized innovation mean decays geometrically at rate $\rho = \varphi(1-K)$ to $\mu_\infty = \delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F})$, where K is the steady-state Kalman gain and F the innovation variance. If $\mu_\infty < k$, the post-transient innovation CUSUM has negative drift, and the probability of an alarm in the next L observations is bounded by $(L+1) \cdot \exp(-2(k-\mu_\infty)h)$ for threshold h.

The detector therefore fires during the adaptation transient or, with exponentially small probability, never. With $\varphi = 0.95$ and $k = 0.5$, $\mu_\infty > k$ would require shifts of order 10σ : the innovation CUSUM is *structurally* in the fast-or-never regime, which is why it is the speed champion but never the detection champion.

Proposition 2 (raw-CUSUM delay). The raw-Y CUSUM sees the full shift as a sustained standardized drift $\Delta = \delta/\sigma_Y$, giving an Albert-Wald mean delay of approximately $h/(\Delta - k)$ once $\Delta > k$, and negligible power when $\Delta \leq k$.

Verification (`experiments/exp06_theory_check.py`, 1000 reps): the Monte Carlo innovation path matches μ_t within MC error; the Proposition 1 bound is never violated; at the actual calibrated thresholds, μ_∞ sorts every observed detection rate — $\delta \leq 1\sigma$ gives bound $\leq 0.7\%$ (observed $\approx \text{FAR}$); $\delta = 3\sigma$ is

a knife-edge ($\mu_\infty = 0.43\text{--}0.48$ vs $k = 0.5$) matching the observed $0.55\text{--}0.67$. Proposition 2's Wald delays 68/84/110 across SNRs bracket the observed raw medians 58/75/91 ($\approx 15\text{--}20\%$ conservative) and explain the partial 1σ detection (0.30) without any fitting: $\Delta = 0.577$ barely exceeds k , so the Wald delay (1334) dwarfs the 250-observation horizon.

The persistence boundary (φ sweep). Proposition 1's regime is not unconditional — $\mu_\infty = \delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F})$ is *increasing in* $(1-\varphi)$, so the fast-or-never trap should loosen as the state becomes less persistent. Sweeping $\varphi \in \{0.5, 0.8, 0.95, 0.99\}$ at fixed SNR (Figure `grid_v6_muinf_scatter.png`) confirms it and turns the theory into a falsifiable ordering: μ_∞ sorts the innovation-CUSUM detection rate across all cells (Spearman 0.94), fast-regime cells ($\mu_\infty \geq k = 0.5$) detect $0.83\text{--}1.00$ while never-regime cells ($\mu_\infty < 0.5$) detect $0.07\text{--}0.67$, and at 3σ the detector escapes the trap at low persistence ($0.98/0.97$ at $\varphi = 0.5/0.8$) but is caught at high persistence ($0.65/0.30$ at $\varphi = 0.95/0.99$) — while the raw CUSUM detects at $0.96\text{--}1.00$ for *every* φ , so the φ -dependence is specific to the filtered detector. The negative first-moment result is therefore a boundary condition of *persistent* latent states — which is the empirically relevant case (trend output, a natural rate, a volatility level are near-unit-root), and the case the rest of the paper studies at $\varphi = 0.95$. The one off-trend cell — $\varphi = 0.99$, low SNR, where detection (0.63) exceeds what μ_∞ (0.21) predicts — is honest evidence of the transient's own contribution: at near-unit-root φ the adaptation transient is so long that its accumulated mass fires the CUSUM even when the asymptotic drift is negligible (the "fast" branch, strengthened by a long transient). μ_∞ governs the post-transient tail; total detection adds the transient mass.

Assumptions and estimation error. Propositions 1–2 assume the steady-state filter with *known* parameters. Two facts keep this from being a limitation in practice. First, the filter reaches its steady state within the 125-observation training prefix, well before monitoring begins (the innovation autocorrelation is flat by then). Second, the error from estimating (φ, q, r) rather than knowing them is second-order: in the M1 equivalence check (§5, `experiments/exp07`) the ARIMA and Kalman standardized-innovation series computed with *estimated* parameters correlate at $\bar{\rho} = 0.99$ with each other and, computed with *true* parameters, at $\rho = 1.000$ to a max discrepancy of $\approx 10^{-9}$ — so the entire estimated-vs-true gap is the small residual that leaves the median correlation at 0.99. The theory describes the estimated filter to within that gap.

5. Second moments: a whitening ladder

Is the latent layer's second-moment advantage about the *state estimate*, or merely about *prewhitening* the autocorrelated observations before applying a variance statistic? We answer this directly by running the identical variance CUSUM (up-arm Page CUSUMs of z^2-1 at allowances $k = 0.25$ and 0.05 , a down-arm CUSUM of $1-z^2$, max over arms, no per-time standardization) at three levels of prewhitening, calibrated by the same routine on the same matched-null seed blocks:

- raw — z from the raw observations, standardized by frozen training-prefix moments;
- ARIMA — the same statistic on the standardized one-step residuals of an AIC-selected, training-prefix-frozen ARIMA model (whitened, but not state-aware);
- latent — the composite's e^2 -based variance features on the Kalman innovations (state-aware).

The ladder has three rungs but only two are distinct. For the AR(1)

- noise DGP the observable Y has an *exact* ARMA(1,1) reduced form: differencing by $(1 - \varphi L)$ leaves an MA(1), whose invertible root gives an MA parameter θ and innovation vari-

ance σ_ε^2 satisfying two identities we verify to machine precision (Appendix B; `lsc.theory.arma11_representation`) — $\sigma_\varepsilon^2 = F$, the Kalman innovation variance, and $\theta = \rho = \varphi(1 - K)$, the Proposition-1 decay rate. So the steady-state Kalman innovations *are* the ARMA(1,1) innovations: the ARIMA and latent rungs are the same filter, not two competitors. We confirm this numerically (`experiments/exp07_arma_equivalence.py`, ≥ 200 null paths): with true parameters the two standardized-innovation series agree to a median correlation of 1.000 (max discrepancy $\approx 10^{-9}$), and with each rung fit on the training prefix — the actual operating condition — they still correlate at $\bar{\rho} = 0.99$, the small wedge being pure estimation error (forcing the ARIMA order to the true (1,0,1) tightens it to 0.9995). (AIC, incidentally, rarely selects (1,0,1): near the unit root at $\varphi = 0.95$ it prefers (1,0,0) or a differencing (0,1,1); this is a benign near-unit-root artifact — those orders approximate the ARMA(1,1) closely enough to preserve $\bar{\rho} \geq 0.95$ — reported in full in Appendix B.) The practical consequence: the ladder is really raw vs. whitened, and “does the *state estimate* help beyond ARIMA whitening?” has the answer *no, by construction*. What remains is the genuinely empirical question — when does whitening help at all? — which turns out to depend on *which variance channel* breaks.

The ladder, both channels (detection rate at $T = 500$, 5% calibrated FAR; MC SEs ≤ 0.02 in `paper_assets/ladder_table.csv`, `break_channel` column).

channel	break	rung	SNR 0.1	SNR 0.5	SNR 2.0
r (obs-noise)	$\times 1.5$	raw	1.00	0.56	0.10
		ARIMA	0.90	0.94	0.87
		latent	0.82	0.87	0.91
	$\times 3$	raw	1.00	1.00	0.85
		ARIMA	0.98	1.00	1.00
		latent	0.99	0.99	0.98
q (state-innov)	$\times 1.5$	raw	0.09	0.21	0.23
		ARIMA	0.03	0.10	0.16
		latent	0.06	0.11	0.23
	$\times 3$	raw	0.72	0.96	0.96
		ARIMA	0.26	0.79	1.00
		latent	0.44	0.76	0.98

Reading the ladder: the ordering inverts across channels. The subtle $\times 1.5$ break is the discriminating case, and it tells opposite stories in the two channels.

Observation-noise (r) breaks — prewhitening wins. The raw rung is strongly SNR-dependent, falling monotonically from 0.996 (SNR 0.1) through 0.560 to 0.102 (SNR 2.0, within 5 pp of the 6.0% empirical FAR, i.e. chance). The mechanism is transparent: as SNR rises the latent state’s variance dominates the marginal variance of Y , so a $\times 1.5$ change in the (now-small) observation-noise component is a shrinking fraction of total variance, masked by state-driven autocorrelation. Prewhitening removes exactly that autocorrelation: the ARIMA rung is flat across SNR (0.90 / 0.94 / 0.87) and tracks the latent rung step for step (0.82 / 0.87 / 0.91) — as it must, since the equivalence above makes them the same filter. On this channel the advantage over raw is *prewhitening under autocorrelation*, decisive where the latent signal masks the noise change.

State-innovation (q) breaks — raw wins, and the SNR-dependence flips. Here the raw rung’s dependence on SNR *reverses sign* — it *rises* with SNR (0.09 $\rightarrow$ 0.21 $\rightarrow$ 0.23 at $\times 1.5$) — and the raw rung matches or beats the whitened rungs at every SNR ($\times 1.5$ ARIMA 0.03 / 0.10 / 0.16; $\times 3$ raw 0.72 / 0.96 / 0.96 vs ARIMA 0.26 / 0.79 / 1.00, the whitened rung catching up only at the $\times 3$ high-SNR ceiling). The mechanism is the mirror image: a q-break inflates the *state’s own* variance, which dominates the marginal

variance of Y at high SNR, so a raw z^2 statistic sees it directly — whereas prewhitening *strips out* the state-carried signal along with the autocorrelation. Prewhitening reveals a break in the white component and removes a break in the state. Quieting ($\times^{2/3}$, i.e. reduced q) is undetectable by every rung (≤ 0.07 , at FAR): a low- q state contributes too little to Y to register its own reduction.

This resolves the pre-registered decision rule (experiments/CHANGELOG.md) as Outcome B2: the “prewhitening beats raw” result is *specific to the observation-noise channel*. It matters because the events the application targets — the Great Moderation, crisis volatility — are state-innovation (shock-variance) breaks, not observation-noise breaks; on that channel a raw variance CUSUM is at least as good as any amount of whitening. This is not a weakness to hide but the explanation for a real-data fact (§9): the raw variance CUSUM’s crisis timing is *indistinguishable* from the state-aware composite’s, exactly as the q -channel predicts. (The earlier r -channel-only framing recorded a provisional “Outcome C”; the q channel is what disambiguates it.) Attribution: the z^2/e^2 variance-pressure arms drive the alarms at every rung.

Runway (T sweep, $\times 1.5$, SNR 0.5). The subtle-break detection scales with sample length for every rung: at $T = 200$ all three are weak (raw 0.26, ARIMA 0.10, latent composite 0.11) — the whitened rungs need runway to accumulate the CUSUM — and by $T = 2000$ all reach ceiling (raw 0.98, ARIMA ≈ 1.00 , latent 0.99). The $T = 200$ calibration is *not* hot for these variance detectors (empirical FAR 5.0–5.8% vs the 5% target), unlike the level CUSUMs (§8.1). At the coarse $\times 3$ break the composite is at ceiling even at $T = 200$ (1.00, median delay 26 of the ~ 100 post-break observations, best level-oriented benchmark 0.17).

Scale *quieting* ($\times^{2/3}$) inverts the up-break pattern and is treated in §6 and §8.3: there the raw rung catches the low-SNR case (0.32 at SNR 0.1, where noise dominates) while the whitened rungs and composite sit at chance, and only the standalone exceedance detector recovers it at moderate SNR.

6. Dynamics: near the information floor

Pure persistence changes (marginals preserved) are close to undetectable: at SNR 0.5, no method exceeds FAR (a pre-registered hypothesis wrong in both directions — raw CUSUM even reached 0.16 on persistence-up via a conditional “level-freeze” artifact we dissect). Quieting changes (φ down) actively *suppress* every excursion statistic below its null level. Purpose-built quietness features (CUSUM of $1 - e^2$, rolling innovation autocorrelation) rescue detection only where the information exists: 0.33 at SNR 2.0 (the only above-FAR persistence detection anywhere in the grid) and 0.17 at $T = 2000$. Scale-quieting ($\times^{2/3}$) is detectable — but, outside the noise-dominated regime, only by the exceedance detector (§8.3): at SNR 0.5 it reaches 0.41/0.33 (Gaussian/ t_5) while every other method sits at chance. The one exception is the same low-SNR window that lifts the raw rung on the up-breaks (§5): at SNR 0.1 the raw variance CUSUM catches the quieting too (0.32), because when observation noise dominates a scale change is directly visible in raw z^2 regardless of its sign.

7. Multiple breaks: everyone is one-shot

Under a re-arm protocol applied identically to all methods (re-arm when the score drains below half threshold plus a 20-obs refractory), raw CUSUM never detects a second event (recall 0.00 — its fixed-baseline statistic saturates and cannot drain), but the LSC CUSUMs rarely re-arm in time either at 150-observation spacing (level $\rightarrow$ level second-event recall ≤ 0.05): first-alarm delay plus drain time exceeds the gap. The exception is cross-channel pairs: level $\rightarrow$ variance is caught by the composite alone

(second-event recall 0.60, F1 0.63) because its variance features were never saturated by the level event. Re-arming costs almost nothing under the null ($\leq 1.2\%$ of null paths give a second alarm) — saturation, not chatter, binds. Bounded-memory (windowed) statistics are the identified fix and are left to future work.

8. Robustness

8.1 Sample size. §5 numbers; short-T caveat: heavy-tailed null maxima make quantile thresholds noisy, and at $T = 200$ the two baseline CUSUMs calibrate hot (8.8–9.4% vs the 5% target). Quote empirical FARs alongside any short-sample power claim.

8.2 Misspecification. t_5 observation noise: rankings preserved; raw CUSUM unhurt on levels (0.99); the composite's subtle-variance case collapses ($\times 1.5$: $0.87 \rightarrow 0.16$) — repaired in §8.3. The raw and ARIMA variance rungs are far less tail-fragile than the composite: on the $\times 1.5$ break at SNR 0.5 the raw rung falls only $0.56 \rightarrow 0.43$ and the ARIMA rung $0.94 \rightarrow 0.74$ under t_5 (against the composite's $0.87 \rightarrow 0.16$), because the plain max-over-arms z^2/e^2 statistic without per-time composite standardization retains the tail excursions that carry the variance signal — the same logic that motivates the standalone exceedance detector (§8.3). At $\times 3$ all variance rungs stay at ceiling under t_5 (raw and ARIMA 1.00, composite 0.97). Nonlinear tanh drift (mildly bimodal state): level detection collapses for *every* method (raw 0.15 at 3σ — the state's own regime-hopping inflates all null thresholds), while variance detection is untouched (0.97 at $\times 1.5$). The two detection families read disjoint information channels.

8.3 Heavy tails and the exceedance repair (a three-act story worth telling honestly). (i) Huberizing e^2 (clip at $2.5 \cdot \text{MAD}$) — falsified, worse everywhere, even Gaussian ($\times 1.5$: $0.87 \rightarrow 0.06$): the variance signal lives in the tail the clip removes. (ii) An exceedance-indicator CUSUM (count of $|e|$ above its training 90th percentile; bounded summand under any distribution) — the raw statistic separates the null and break classes near-perfectly. Dropped into the composite in place of the e^2 -pressure features (the "robust2" variant), and standardized per-time-point as in §8.4, it is *diluted, not dead*: it reaches 0.58 at $\times 1.5$ (SNR 0.5, Gaussian) and 0.21 at $\times 1.5$ under t_5 — where it even edges the e^2 -based composite's heavy-tail collapse (0.16, §8.2) — and ≈ 0.97 at $\times 3$ under t_5 . But it trails the standalone detector of (iii) at the subtle break, because a max-over-features composite spreads the calibrated 5% false-alarm budget across all features: the exceedance feature's per-t standardized score climbs past $z \approx 19$ on a $\times 1.5$ break, so bounded increments *do* reach discriminating ratios (an earlier pooled-scale build, before the §8.4 fix, wrongly suggested otherwise), yet the shared threshold still costs it power relative to calibrating the statistic alone. The remedy is *exposure*, not a different feature. (iii) The same statistic as a *standalone* calibrated detector (up-arm $k = 0.05$, down-arm $k = 0.02$; k chosen on non-evaluation seeds, procedure logged): variance $\times 1.5$ at 0.87 Gaussian / 0.75 t_5 (repairing 0.16), $\times 3$ at ~ 1.0 with ~ 37 -obs delay under both distributions, and the first successful quieting detection ($\times 2/3$: 0.41/0.33). Both headline rates fell 3–5pp short of the pre-registered bars — reported as such.

8.4 The local-level (random-walk state) arena: degenerate for levels, whitening-mandatory for variances. The canonical latent-state model in economics is the local level (random-walk state); we ran the level and variance ladder cells there rather than dismiss it (configs/grid_v7_l1level.yaml, Figure grid_v7_l1level_degeneracy.png), and it splits cleanly. *Level* detection is degenerate for every method: at 3σ all five detectors sit at the 5% FAR (raw CUSUM 0.07–0.10, Kalman-innovation 0.04–0.15, raw/ARIMA variance and composite ≤ 0.07). A level break in a random-walk state is absorbed by a well-specified filter as one large ordinary innovation — no sustained signal — and the raw-Y CUSUM has no fixed baseline: its calibrated threshold is 1500–2400 (versus $O(10\text{--}100)$ in the AR(1) arena) and

it still calibrates hot. There is no common null against which to rank level detectors, which is why we study the identifiable AR(1) arena in the body. But *variance* detection is not degenerate — it merely requires whitening. On a $\times 1.5$ observation-noise break the raw z^2 CUSUM is at chance (0.06 at every SNR; threshold $\approx 10^4$, meaningless on a nonstationary series) while the ARIMA-differencing rung detects (0.71 / 0.84 / 0.58) and the Kalman composite detects (0.97 / 0.89 / 0.68); at $\times 3$ the whitened rungs reach 0.97–1.00 while raw stays at chance. This is the exact complement of §5’s r-channel AR(1) result — where a raw variance CUSUM could *win* when observation noise dominates — and it sharpens the paper’s thesis: prewhitening is not merely helpful but *mandatory* once the observable is nonstationary, because the raw statistic has no stationary baseline to calibrate against.

Protocol lessons (each cost us a wrong result before it was fixed). plain-HMM regime probabilities saturate and cannot be FAR-calibrated on nonstationary data; probability-scale scores need log-odds; EM needs persistent-initialization restarts; composite features must be standardized per-time-point, not pooled; order-statistic thresholds have Beta($n+1-k$, k) noise regardless of distribution, so heavy-tailed detectors need larger calibration budgets.

9. Real data (illustrative)

Three FRED series, pinned snapshots (2026-07-11), rolling causal monitoring (train 120 months / monitor 60), per-segment parametric bootstrap calibration at 5% FAR per window, alarms attributed to the feature that crossed.

Industrial production (INDPRO, 1948–2026). Composite alarms: 2008-09 and 2020-04 (both *variance_pressure*), 1990-12 (*variance_quiet*), 1969-08 (*variance_quiet*; within the false-alarm budget and reported as such). Hits 3/9 NBER peaks within 12 months, 1 stray vs 0.7 expected; permutation $p = 0.007$ (innovation CUSUM $p = 0.018$; raw CUSUM 1 hit, $p = 0.15$). The raw variance CUSUM — the bottom rung of the ladder, added here to test real-data uniqueness — does catch the GFC (2008-09, up-arm, the same month as the composite) but embeds it among four stray quieting alarms (1967, 1968, 1988, 2019) and misses COVID, so its NBER association is not significant (5 alarms, 1 hit, $p = 0.56$). The crisis *is* detectable by a raw variance statistic; what the latent layer buys on real data is a *clean* alarm profile — a significant, low-stray association — not the crisis catch itself.

What real-time data changes (ALFRED vintages). Re-running each decision on the data *as it existed that month* both tempers and sharpens the timing claims. COVID: robust — the composite’s real-time alarm is at the 2020-04 vintage (data month 2020-03), one month earlier than on revised data and ~ 2 months before the NBER announcement. GFC: downgraded — the crossing is at the same data month (2008-09) but only in the 2008-12 vintage, so real-time knowledge is coincident with the NBER announcement (2008-12-01), though still 4 months ahead of the raw level CUSUM’s real-time alarm (2009-04). Because the raw variance CUSUM alarmed on revised INDPRO, we ran it through the identical vintage protocol for an apples-to-apples comparison: its real-time timing is *indistinguishable from the composite’s* — same vintage and same data month for both GFC (2008-12 vintage, 2008-09) and COVID (2020-04 vintage, 2020-03), both ahead of the raw level CUSUM. On real-data crisis *timing*, then, the prewhitening rung and the state-aware composite are interchangeable; the composite’s advantage is confined to the clean association profile of §9 and to the simulation-calibrated $\times 1.5$ subtlety threshold of §5, not to real-time crisis detection.

GDP (GDPC1, quarterly). GFC and COVID caught (*variance_pressure*, 2008Q4 and 2020Q2). The raw variance CUSUM again catches both crises (2009Q2 and 2020Q2, up-arm) with fewer alarms than on INDPRO (2 alarms, 1 hit, $p = 0.31$) — the same “catches the crisis, misses the clean association” pattern.

The registered Great Moderation event (1984Q1) is an honest miss: causal rolling monitoring detects the quieting only in 1992Q2 (composite) / 1997Q1 (tail shortfall), because 60-quarter training windows contain 1970s volatility until the early 1990s. Retrospective full-sample break dates are not reproducible by honest monitoring.

10-year Treasury yield changes (GS10). Volcker regime caught 4 months after the registered 1979-10 event (variance_pressure; raw and innovation CUSUM the same month); the post-Volcker disinflation quietings are flagged (1989–1996, shortfall/variance_quiet); the 2008 ZLB event is missed within 12 months. The raw variance CUSUM also flags Volcker (1980-02, up-arm) but is the noisiest detector on this series (9 alarms, 8 stray, $p = 0.28$) — on a series that is *all* volatility regime, a raw z^2 statistic fires on every regime shift, which is exactly why its event-association washes out.

Sensitivity. FAR = 10% behaves as expected. Training window 180 months breaks both variance-based detectors' bootstrap calibration on nonstationary real data — the composite and the raw variance CUSUM each fire 14 alarms / 21 windows — while the level (raw CUSUM, 2), innovation (3), and tail (5) detectors stay sane. It is the *second-moment* statistic, not the composite machinery, that is sensitive to a training window long enough to straddle a volatility regime; training windows must be short enough to be locally stationary (120 months worked).

10. Discussion

Four points summarize what the harness bought us.

- The latent-state diagnostics layer is not a better level-shift detector; it is a *different instrument*, reading second moments and dynamics. But the whitening ladder (§5) sharpens the claim past the point where "the state estimate" survives as an ingredient: on this DGP the Kalman innovations *are* the ARMA(1,1) innovations (an exact reduced-form equivalence, verified to machine precision), so whatever second-moment power the latent layer has is prewhitening, full stop — ARIMA residuals capture it identically. What the state framing still buys is dynamics features (persistence, quieting) and the fast-or-never speed edge on levels. The division of labor is now measured (grids), reduced (raw vs. whitened, the two whitened rungs proven identical), and derived (fast-or-never with its ϕ boundary).
- Practical recipe, now channel-aware. Run raw CUSUM for levels. For scale changes the right move depends on *which variance moves*, and one usually does not know: (i) if the break is in the observation (measurement) noise, *whiten first* (ARIMA residuals suffice; the Kalman state adds nothing) — prewhitening is decisive when the series is autocorrelated and the latent signal is strong, and a raw variance CUSUM works only when observation noise dominates; (ii) if the break is in the state's own shock variance — the Great-Moderation / crisis- volatility case — a raw variance CUSUM is at least as good as whitening at every SNR, and whitening can *hurt*. Because the practitioner rarely knows the channel, running *both* a raw and a whitened variance CUSUM (cheap, and they agree on the q channel) is the robust default. Use the exceedance-indicator variant under heavy tails and the composite for breadth. Calibrate everything on matched nulls at a common FAR (a common ARL_0) and report empirical FARs.
- The calibrated-parity protocol itself is a contribution: it exposed every failure mode above, and it is what makes the negative results informative rather than anecdotal.
- Limitations / future work: bounded-memory statistics for multiple breaks; adaptive composite weighting (breadth tax); switching-SSM (Kim filter) model layer; formalizing the persistence-break mechanisms; a vol-regime reference set for scoring the exceedance detector on real data.

Appendix A. Reproducibility

`make all` regenerates every table and figure from pinned seeds (Python 3.14, `statsmodels/hmmlearn`; `make fred` / `make realdata` / `make realtime` for the data applications, snapshots under `data/`). The referee-hardening round adds five reproducible artifacts to the pack: `exp07` (ARMA equivalence, M1), `grid_v5` (the q -break channel, M2), `grid_v6` (the ϕ sweep, M3), `grid_v7` (the local-level arena, M4), and `ar1` (ARL₀/ARL₁ table, M5); all are pinned-seed and join the existing grids draw-for-draw. 95 tests include bit-identical no-lookahead checks for every feature and detector (including the raw and ARIMA variance rungs and a training-freeze check), DGP ground-truth checks (now including the state-innovation break's SD scaling and null-match), the ARMA(1,1)/Riccati identity and Kalman-equivalence guards, calibration- parity checks, and composite golden-score regression guards (which pin the per-time-point-standardized output so a stale artifact cannot recur). All post-hoc design changes and pre-registered hypotheses (including the three falsified ones and the two rejected robust-feature designs — one falsified, one diluted, §8.3) are in `experiments/CHANGELOG.md`; full experiment narratives in `experiments/FINDINGS.md`; theory derivations in `experiments/THEORY.md` (Appendix B). The complete source, pinned data snapshots, and seed configuration constitute the replication package, to be posted publicly on acceptance and available from the author on request in the interim.

Appendix B. Theory derivations

Full derivations of Proposition 1 (the geometric decay of the standardized innovation mean to μ_∞ and the resulting fast-or-never alarm bound) and Proposition 2 (the Albert–Wald delay of the raw-Y CUSUM under a sustained standardized drift) are in `experiments/THEORY.md`, with the numerical verification in `experiments/exp06_theory_check.py` (1000 replications). The steady-state Kalman gain K and innovation variance F are the fixed-point solutions of the scalar Riccati recursion for the AR(1) state model of §2; μ_∞ , ρ , and the bound follow by taking the post-break innovation as a deterministic geometric transient plus mean-zero noise and applying a one-sided Hoeffding bound to the CUSUM increments.

ARMA(1,1) equivalence of the whitened rungs (§5). Applying the AR operator to the observable, $(1 - \phi_L)Y_t = w_t + v_t - \phi v_{t-1} =: u_t$, which is an MA(1) with autocovariances $\gamma_u(0) = q + r(1 + \phi^2)$, $\gamma_u(1) = -\phi r$, $\gamma_u(h) = 0$ ($h \geq 2$). Matching u_t to $(1 - \theta_L)\varepsilon_t$ (variance σ_ε^2) gives the invertible root $\theta = (m - \sqrt{m^2 - 4})/2$ with $m = (q + r(1 + \phi^2))/(\phi r)$, and $\sigma_\varepsilon^2 = \phi r / \theta$. Two identities close the loop with Proposition 1: $\sigma_\varepsilon^2 = F$ and $\theta = \rho = \phi(1 - K)$. The second follows from the innovation recursion $v_t = Y_t - \phi Y_{t-1} + \phi(1 - K)v_{t-1}$ (substitute $\hat{S}_{t-1} = \phi \hat{S}_{t-2} + K v_{t-1}$ into $v_t = Y_t - \phi \hat{S}_{t-1}$), which is the ARMA(1,1) innovation recursion once $\theta = \phi(1 - K)$. Hence the steady-state Kalman innovations and the ARMA(1,1) innovations are the same series; the state layer is not a distinct information set from ARIMA whitening on this DGP. Both identities hold to $< 10^{-12}$ (`lsc.theory.arma1_representation`, `test_arma1_riccati_identities`), and the numerical equivalence on null paths — with true and with estimated parameters — is in `experiments/exp07_arma_equivalence.py` (§5). A near-unit-root caveat: AIC over the benchmark order grid rarely selects the exact (1,0,1) at $\phi = 0.95$ (it prefers (1,0,0) at low SNR, the differencing (0,1,1) at higher SNR), but these approximate the ARMA(1,1) closely enough that the median innovation correlation with the Kalman filter stays at 0.99; forcing (1,0,1) restores 0.9995.

Appendix C. Summary of key quantities

Claim	Number	Source
Raw CUSUM level 3σ detect, all SNRs	0.97–0.99	grid_v1
Innovation CUSUM delay vs raw, 3σ	24–53 vs 58–91 obs	exp02
Innovation CUSUM detect at 3σ	0.55–0.67	exp02/grid_v1
$\mu\infty$ at 3σ , SNR 0.5 (knife-edge vs $k=0.5$)	0.469	exp06
Composite variance $\times 1.5$, $T=500$	0.82 / 0.87 / 0.91 (SNR 0.1/0.5/2.0)	grid_v1
Ladder $\times 1.5$: raw rung, $T=500$	1.00 / 0.56 / 0.10 (SNR 0.1/0.5/2.0)	grid_v4_varbench
Ladder $\times 1.5$: ARIMA rung, $T=500$	0.90 / 0.94 / 0.87 (SNR 0.1/0.5/2.0)	grid_v4_varbench
Ladder $\times 1.5$ t_5 (raw/ARIMA/composite), SNR 0.5	0.43 / 0.74 / 0.16	grid_v4_varbench, grid_v2_missp
ARMA $\equiv$ Kalman innovation $\bar{\rho}$ (estimated / true params)	0.99 / 1.000 ($\max \Delta \approx 10^{-9}$)	exp07 (M1)
q-break $\times 1.5$: raw rung, $T=500$	0.09 / 0.21 / 0.23 (SNR 0.1/0.5/2.0)	grid_v5_qbreak (M2)
q-break $\times 1.5$: ARIMA rung, $T=500$	0.03 / 0.10 / 0.16 (SNR 0.1/0.5/2.0)	grid_v5_qbreak (M2)
Decision-rule outcome fired	r-channel provisional C, resolved B2 (q inverts)	CHANGELOG (M2)
φ sweep: $\mu\infty$ sorts innovation-CUSUM detection	Spearman 0.94	grid_v6_phisweep (M3)
Innovation CUSUM 3σ escape at low φ ($\varphi=0.5$ / 0.99)	0.98–1.00 / 0.30–0.63	grid_v6_phisweep (M3)
Local-level 3σ level detect (all methods)	≤ 0.15 ($\approx$ FAR)	grid_v7_llevel (M4)
Local-level $\times 1.5$ variance (raw / ARIMA rung)	0.06 / 0.58–0.84	grid_v7_llevel (M4)
ARL ₀ at 5% window-FAR ($L=375$)	≈ 7300 obs	arl_table (M5)
Variance $\times 1.5$ across $T = 200/500/2000$	0.11 / 0.87 / 0.99	grid_v2_T
t_5 collapse and repair ($\times 1.5$)	0.16 $\rightarrow$ 0.75 (tail_cusum)	grid_v2_misspec, grid_v3c
Quieting $\times 2/3$ (only tail_cusum)	0.41 / 0.33	grid_v3c
Persistence-down, best anywhere	0.33 (SNR 2.0)	grid_v1
Multi-break: raw second-event recall	0.00	exp04
Composite level $\rightarrow$ var second event	0.60 (F1 0.63)	exp04
INDPRO permutation p (composite)	0.007	rd_eval
GFC real-time	data month 2008-09, known 2008-12	rd_realtime
COVID real-time	data 2020-03, ~ 2 mo before NBER	rd_realtime